



**STANDARD
100**



**ORGANIC
COTTON**

Testing methods

OEKO-TEX® STANDARD 100 & OEKO-TEX® ORGANIC COTTON

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OEKO-TEX®

International Association for Research and Testing in
the Field of Textile and Leather Ecology

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General remarks

This document informs about the analytical methodology applied for chemical safety testing in all laboratories of the OEKO-TEX® Association. The methods described below are applied to any article certified under the certification schemes OEKO-TEX® STANDARD 100 or OEKO-TEX® ORGANIC COTTON. If an article shall be certified according to these standards and also contains leather and skins (with or without hair) or chemicals (e.g. gel pouches), these materials are tested according to the conditions and criteria of OEKO-TEX® LEATHER STANDARD or OEKO-TEX® ECO PASSPORT. For these methods, please refer to the [corresponding testing method documents](#).

Abbreviations

- AAS - atom absorption spectrometer
- CI - chemical ionisation
- DAD - diode array detector
- EI - electron impact
- FLD - fluorescence detector
- IC - ion chromatography
- ICP - inductively coupled plasma
- GC - gas chromatography
- LC - liquid chromatography
- MS - mass spectrometry
- OES - optical emission spectroscopy
- PCR - polymerase chain reaction
- UV/VIS - ultraviolet-visible



pH value (M-01)

The pH value is determined according to ISO 3071 (KCl solution).

Qualitative formaldehyde analysis (M-48)

Samples are extracted in hot water and formaldehyde is derivatised with acetyl acetone and ammonium acetate. The yellow colouration formed is visually assessed under standardized conditions. Similarly, a reaction of formaldehyde with chromotropic acid disodium salt and sulfuric acid, yielding violet colouration, can be utilized.

Quantitative formaldehyde analysis (M-08)

To determine free or releasable formaldehyde, samples are prepared according to Japanese Law 112 / JIS L 1041 - 2011 (using acetylacetone, method B) and DIN EN ISO 14184-1. Analysis is performed by UV/VIS spectroscopy or LC-FLD.

Extractable heavy metals (M-10)

Heavy metals are extracted with artificial acidic sweat solution according to DIN EN 16711-2. The extract is analysed by ICP-OES, ICP-MS or AAS. Metallic accessories with a surface finish or coating are subjected to an additional test for extractable nickel after a pretreatment (wear and corrosion according to EN 12472:2020, used for its abrasion medium).

Chromium VI (M-12)

Chromium is extracted by use of artificial acidic sweat solution according to DIN EN 16711-2. The content of chromium (VI) in the extract is determined with selective and sensitive methods. The detection can be carried by UV/VIS spectroscopy using 1,5-diphenylcarbazide (according to EN ISO 17075-1), AAS, ICP or IC (according to DIN EN ISO 17075-2).

Total heavy metals (M-21)

Analysis of total heavy metal and silver content is based on the method based on requirements defined in DIN EN 16711-1 and CPSC-CH-E1001-08.3 / CPSC-CH-E1002-08.3. Samples are chemically digested using acids to obtain a clear extract containing heavy metals, which is afterwards analysed by ICP-OES, ICP-MS or AAS. Different sample components, which can be differentiated macroscopically (base material, paints, etc.), are separately analysed. This allows to analyse total lead content according to requirements of the American legislation for children's articles (CPSIA, Consumer Product Safety Improvement Act).

Polar and apolar pesticides (M-06)

Polar and apolar pesticides are extracted by ASE (or Soxhlet) using methanol or acetone, or using a QuEChERS extraction. After clean-up, extracts are analysed for pesticides with GC-MS and LC-MS.

Glyphosate and salts (M-06-C)

Glyphosate and its salts are extracted from samples by aqueous extraction, followed by derivatisation or without derivatisation, according to DIN EN ISO 16308 with slight modifications. Extracts are then analysed by LC-MS.

Chlorinated phenols and ortho-phenylphenol (M-07)

The samples are extracted with a basic aqueous solution following DIN EN 17134-2. The extracted free phenols and possibly hydrolysed phenol esters are acetylated, transferred to an organic phase and analysed with GC-MS.



Phthalates & siloxanes (M-18)

Analysis of phthalates follows the procedure in DIN EN ISO 14389, and siloxanes are also analysed using this method. Extraction of the testing material with tetrahydrofuran is followed by precipitation of the polymers with (cyclo)hexane. The extract is analysed by GC-MS.

Organic tin compounds (M-17)

The method is based on DIN EN ISO 22744-1 with modifications. An extraction of sample with an ethanol/acetic acid solution and tropolone is followed by derivatisation with sodium tetraethyl borate. The extract is analysed by GC-MS.

Bisphenols & related substances (M-40)

The determination of bisphenols & related substances is performed by extraction of samples with THF in an ultrasonication bath followed by a polymer precipitation with methanol. The extract is then analysed by LC-MS.

Azo dyes, arylamines and aniline (M-03)

Tests for azo dyes, which may be cleaved into carcinogenic arylamines, are carried out according to ISO 14362-1 and 14362-3 and are combined with the analysis of aniline (both as free chemical residue and cleavable form within colourant). Samples are treated with sodium dithionite and extracts are analysed with two chromatography methods, preferably LC-MS and GC-MS.

Allergenic, carcinogenic and other banned colourants & quinoline (M-04)

The identification and quantification of dyestuff with allergenic or carcinogenic potential, other banned dyestuff and pigments and Michler's ketone and base is achieved by hot acetone extraction and analysis by LC-DAD or LC-MS.

Chlorinated benzenes and toluenes (M-02)

The method is based on DIN EN 17137, with an ultrasonication bath extraction of the testing materials with dichloromethane. The extracts are analysed by GC-MS.

Polycyclic aromatic hydrocarbons (PAH; M-23)

The method is based on DIN EN 17132, with an extraction of the test samples with toluene. The extracts are analysed with GC-MS.

Flame retardants (M-30)

The determination of banned flame retardants (brominated, organophosphorus and others) is based on sample extraction with a mixture of methanol and toluene, followed by GC-MS and LC-MS analysis, respectively.

Solvent residues (M-26)

The method is based on an extraction of the test samples with methanol. The extracts are analysed with GC-MS.



Alkylphenols & alkylphenol ethoxylates (AP & APEO; M-25)

The method is based on DIN EN ISO 18254-1 with modifications. After extraction of the test samples with methanol in an ultrasonication bath. The extracts are analysed with LC-MS.

Per- and polyfluoroalkyl substances (PFAS; M-22)

PFAS analysis is carried out in accordance with DIN EN 17681-1:2025. PFAS are extracted from the samples using strongly alkaline methanol in an ultrasonication bath. This alkalinity allows for hydrolysis of, for instance, fluorinated polymers and esters and results in release of PFAS. After neutralization with acid, PFAS are analysed using LC-MS.

Total fluorine (M-42)

The method is based DIN EN 17813 with modifications, ASTM D7359-23 with modifications or ISO 20999 with modification. Samples are combusted with oxygen. The resulting HF is collected in an absorber solution and is analysed for the fluorine content using IC.

UV stabilisers (M-28)

The method is based on ISO 24040, with modifications. After extraction of samples with tetrahydrofuran (THF) and polymer precipitation with acetonitrile, the extracts are then analysed with LC-MS.

Short & medium chain chlorinated paraffins (SCCP&MCCP; M-24)

The method for determination of the short and medium chain chlorinated paraffins is based on an extraction of the testing material with a mix of dichloromethane/(cyclo)hexane, followed by a clean-up and subsequent analysis with GC-MS. For a total analysis (sum of short and medium chained chlorinated paraffins) the instrument is operated in EI mode. CI mode is used for the identification and quantification of SCCP and MCCP congeners present in the sample.

N-nitrosamines and N-nitrosatable substances (M-34)

During this test according to EN 71-12, N-nitrosamines and N-nitrosatable substances migrate into a saliva test solution. The N-nitrosatable substances react to N-nitrosamines by acidification. The analysis of the freely available as well as produced N-nitrosamines is done by LC-APCI-MS.

VOCs, glycols, cresols & chlorinated solvents (M-31)

The sample to be analysed for volatile organic compounds, glycols, cresols, & chlorinated solvents is baked out by thermal desorption. The released substances are enriched on suitable trapping material and analysed by GC-MS.

Vinylchloride (M-50)

The analysis method for vinylchloride follows DIN EN ISO 6401. Monomeric vinylchloride is extracted from samples using DMAc, followed by headspace GC-MS.

Azodicarbonamide (ADCA; M-35)

The sample is extracted with DMSO. After extraction the sample is analysed by LC-DAD.



Dimethylfumarate (DMFu; M-27)

The method is based on DIN EN 17130, with an extraction of the samples with acetone. After solvent evaporation, the extracts are analysed with GC-MS.

Melamine (M-44)

The samples are extracted with water in a shaking bath (according to Japanese Law 112 / JIS L 1041 – 2011) and then analysed on LC-DAD and / or LC-MS.

N-(hydroxymethyl)acrylamide (M-43)

The method is based on extraction of samples with hot water in an ultrasonication bath. The extract is analysed by LC-DAD.

Phenol & related substances (M-39)

The determination of phenol and related substances is performed by extraction of the test material with methanol in an ultrasonication bath. The extract is analysed by LC- FLD additionally equipped with a DAD.

Emission of volatile chemicals (M-14 & M-15)

For the determination of emitted volatile chemicals, the methods are based on ISO 16000-3 (formaldehyde) and ISO 16000-6 (VOCs), using an emission chamber according to ISO 16000-9. For emission of VOCs, a microchamber instead of an emission chamber according to ISO 16000-9 can be used. Formaldehyde is adsorbed on DNPH cartridges, eluted with acetonitrile and analysed using LC-DAD or LC-FLD. For VOC analysis, different adsorbents are used and analysis is performed by thermal desorption and GC-MS.

Colour fastness (M-09)

In all colour fastness tests, only the fastness grades with respect to staining of the adjacent fabrics are determined. The basic methods for the tests are ISO 105-A01 and ISO 105-A03. More specifically, the following tests are performed:

- Determination of colour fastness to water according to ISO 105-E01
- Determination of colour fastness to acidic and alkaline perspiration according to ISO 105-E04
- Determination of colour fastness to rubbing dry according to ISO 105-X12
- Determination of colour fastness to saliva and perspiration. The test is performed with reference to § 64 LFGB (German law for food, commodities and animal feed), DIN 53160, ISO 105- A03 and ISO105-E04

General odour on clothing and accessories (M-16)

All articles are subjected to an initial odour test. If this fails, the certification procedure is stopped. The odour from mould, high boiling fractions of petrol (from colour printing), fish (from permanent finishing) or aromatic hydrocarbons will lead to a test failure. Moreover, odorants (perfumes) used for removing or masking the smell of a textile material originating from its production (oil, fats, dyestuffs) must not be detected during sensory odour testing.

Odour on other articles (M-16)

Some articles not used for clothing (e.g. textile floor coverings, mattresses, foams, and large coated articles) undergo an odour test procedure with reference to SNR 195 651. Such articles may emit a perceptible odour. This is an inherent initial odour typical of the products and normally vanishes after some weeks. For the test, a sample of defined area is



conditioned in a desiccator of set humidity and temperature for a defined time. The odour formed in the closed system is evaluated sensorially by a set of test persons.

Fibre composition analysis (M-45)

Wherever feasible, non-fibrous material is removed from the textiles by suitable pretreatment and the fibres are separated by hand, dried and weighed in order to calculate the proportion of each fibre in the mixture. Microscopic analysis can assist this process, especially to identify cotton. If fibres form a more intimate mixture which does not allow manual separation, chemical analysis with solvents or acids is carried out according to ISO 1833-12, ISO 1833-11 or ISO 1833-6, depending on the textile's fibre composition.

Glitter fastness (M-47)

Adhesive tape is applied to a textile surface with applied glitter particles and pressure is applied by a standardised rubbing motion. The tape is removed and the number of glitter particles in a defined tape area is counted.

Qualitative GMO analysis (M-33-A)

The detection of genetic modification is carried out according to ISO5354, in which DNA is isolated from chemically untreated cotton and analysed for various markers that indicate genetic modification of the cotton. The test allows to screen for all currently known genetically modified cotton events and is designed to optimize the probability of also detecting unknown genetically modified cotton events.

Quantitative GMO analysis (M-33-B)

The procedure is carried out after qualitative analysis. The isolated DNA is further analysed by quantitative real-time PCR. The fundamental cotton lines are determined in the first step and quantified in the second step. The quantification is performed in relation to the taxonomic target gene control (SAH7).